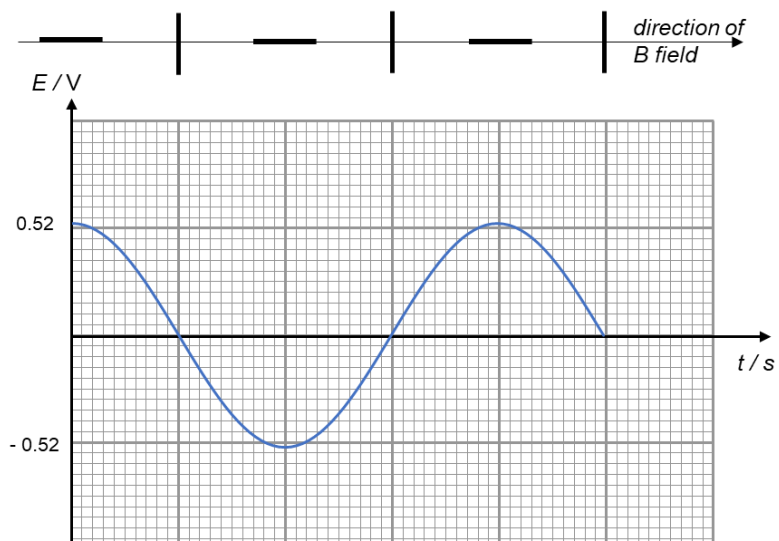


3(a)	The magnetic flux linkage through a whole coil of N turns is the sum of the magnetic flux through the individual turns
(b)(i)	As the coil rotates, the area of the coil perpendicular to the field changes sinusoidally causing the magnetic flux linkage to change sinusoidally and a sinusoidal e.m.f. is induced,
	Since the circuit forms a closed loop, a sinusoidal current will now follow
(ii)	$E = -\frac{d\Phi}{dt} = -\frac{d}{dt}NBA \cos(\omega t) = \omega NBA \sin(\omega t)$
	$E_{\max} = \omega NBA = 2\pi\left(\frac{30}{60}\right)(6.2)(265 \times 100^{-2}) = 0.516 \text{ V}$
	$= 0.52 \text{ V}$
(iii)	orientation of coil seen from the top view 
(c)(i)	$V_{\text{rms}} = \frac{V_0}{\sqrt{2}} = \frac{0.516}{\sqrt{2}} = 0.365 \text{ V}$
(ii)	$\frac{N_s}{N_p} = \frac{V_s}{V_p}$ $0.25 = \frac{V_s}{V_p}$ $V_s = 0.25 \times 0.365 = 0.09125 \text{ V}$
	$I_s = \frac{V_s}{R} = \frac{0.09125}{10} = 9.13 \times 10^{-3} \text{ A}$